1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
KD2_SAURABH_98912>select o.onum,o.amt,o.odate,o.cnum,o.snum from orders o join customers c on o.snum=c
.snum where o.onum in (select onum from orders where c.cname='cisneros');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
2 rows in set (0.00 sec)
```

7. Extract all the orders of Motika.

8. Find all the orders of salespeople servicing customers in London.

```
KD2_SAURABH_98912>select o.onum,o.amt,o.odate,o.cnum,o.snum from orders o join salespeople s on o.snum
=s.snum where city='london';
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
4 rows in set (0.00 sec)
```

6. Write a query that selects all orders for amounts greater than any for the customers in London.

```
KD2_SAURABH_98912>select * from customers where rating > any(select min(rating) from customers where c
ity='rome');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)
```

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres’. (Hint: find all customers of serres salesperson).

```
KD2_SAURABH_98912>select * from customers where rating >= any(select min(rating) from customers where
snum=1002);
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)
```

11. Select customers who have a greater rating than any other customer in Rome.

```
KD2_SAURABH_98912>select * from customers where rating > any(select min(rating) from customers where c
ity='rome');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)
```

12. Select all orders that had amounts that were greater that at least one of the orders from ‘1990-10-04’.

```
KD2_SAURABH_98912>select * from orders where amt >any(select min(amt) from orders where odate='1990-10
-04');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
8 rows in set (0.00 sec)

KD2_SAURABH_98912>
```

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
KD2_SAURABH_98912>select * from orders where amt < (select max(o.amt) from orders o join customers c o
n o.snum=c.snum where c.city='san jose');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
+------+---------+------------+------+------+
8 rows in set (0.00 sec)
```

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
KD2_SAURABH_98912>select * from orders where amt >any(select min(amt) from orders where odate='1990-10
-04');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
8 rows in set (0.00 sec)

KD2_SAURABH_98912>
```

14. Select those customers whose rating are higher than every customer in London

```
KD2_SAURABH_98912>select * from customers where rating > (select min(rating) from customers where city
='london');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)
```

. 5. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
KD2_SAURABH_98912>select * from customers where rating > (select min(rating) from customers where city
='london');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)
```

9. Find names and numbers of all salesperson who have more than one customer.

```
ERROR 1054 (42S22): Unknown column 'cnum' in 'having clause'
KD2_SAURABH_98912>select sname,snum from salespeople where snum in (select snum from customers group b
y snum  having count(cnum)>1);
+--------+------+
| sname  | snum |
+--------+------+
| Peel   | 1001 |
| Serres | 1002 |
+--------+------+
2 rows in set (0.01 sec)
```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
KD2_SAURABH_98912>select o.snum,sum(o.amt) as total_amt  from orders o group by o.snum having sum(o.am
t)>(select max(amt) from orders);
+------+-----------+
| snum | total_amt |
+------+-----------+
| 1001 |  15382.07 |
+------+-----------+
1 row in set (0.00 sec)
```

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
KD2_SAURABH_98912>select s.snum,s.sname,s.city,count(c.cnum) as totalc from salespeople s join custome
rs c on s.snum=c.snum group by s.snum,s.sname,s.city having count(c.cnum)>1;
+------+--------+----------+--------+
| snum | sname  | city     | totalc |
+------+--------+----------+--------+
| 1001 | Peel   | London   |      2 |
| 1002 | Serres | San Jose |      2 |
+------+--------+----------+--------+
2 rows in set (0.00 sec)
```